DriveSafe

Android-Based Driver Drowsiness Detection System

Technical Architecture, Computer Vision and Implementation

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1 Project Overview

1.1 Problem Statement

Driver drowsiness develops gradually and can be observed through changes in facial behaviour. Prolonged eye closure, repeated yawning and forward head movement are common indicators of reduced alertness.

DriveSafe is an Android application that uses the device's front camera to monitor these indicators in real time. Facial landmarks are analysed to estimate eye openness, mouth movement and head position. When sustained signs of drowsiness are detected, the application generates audio and vibration alerts.

The main challenge is distinguishing genuine drowsiness from normal behaviour such as blinking, talking, looking at mirrors and temporary landmark errors. DriveSafe addresses this through temporal tracking, smoothing, threshold-based classification and combined scoring.

1.2 Objectives

The main objectives of the project are:

- Extract facial landmarks using MediaPipe Face Landmarker.
- Detect eye closure and distinguish normal blinks from prolonged closures.
- Identify yawning and head-nodding behaviour.
- Combine multiple indicators into a drowsiness score.
- Generate audio and vibration alerts when required.
- Perform processing locally without relying on a server.
- Handle missing faces, dropped frames and unstable landmarks.

1.3 Technology Stack

Table 1: Technologies used in DriveSafe.

Layer	Technology	Purpose
Platform	Android, Kotlin	Application and detection logic.
Camera	CameraX	Live preview and frame acquisition.
Face detection	MediaPipe Face Landmarker	Facial landmark extraction.
Image conversion	Custom YUV converter	Converts camera frames into ARGB bitmaps.
Processing	Single-thread executor	Background frame analysis.
Alert system	MediaPlayer, ToneGenerator, Vibrator	Audio and haptic notifications.

1.4 System Capabilities

DriveSafe analyses six main sources of information:

- Eye Aspect Ratio (EAR) for eye closure.
- Bilateral eye fusion to reduce one-sided detection errors.
- Closure duration and rolling PERCLOS.
- Mouth Aspect Ratio (MAR) for yawn detection.

- Facial landmarks for approximate head-pitch estimation.
- Face visibility tracking to distinguish missing faces from closed eyes.

The research also describes a per-driver calibration phase and adaptive baseline tracking. These are discussed in Section 9.

1.5 Source Code Structure

The main implementation is organised as follows.

Table 2: Main source files and responsibilities.

Package	File	Responsibility
Root	`MainActivity.kt`	Camera setup, frame processing and UI.
camera	`YuvBitmapConverter.kt`	YUV-to-ARGB conversion.
vision	`FaceLandmarkerHelper.kt`	MediaPipe integration and landmarks.
detection	`EyeAspectRatioCalculator.kt`	EAR calculation.
detection	`MouthAspectRatioCalculator.kt`	MAR calculation.
detection	`EyeClassifier.kt`	Smoothing and eye classification.
detection	`ClosureEpisodeTracker.kt`	Closure timing and classification.
detection	`PerclosCalculator.kt`	Rolling PERCLOS calculation.
detection	`DrowsinessDetector.kt`	Detection pipeline coordination.
detection	`DrowsinessScoringEngine.kt`	Drowsiness score and state.
detection	`HeadPoseEstimator.kt`	Head-pitch estimation.
audio	`AlertManager.kt`	Audio and vibration alerts.

1.6 On-Device Processing

The complete detection pipeline operates inside the Android application. Camera frames are converted, analysed and scored locally.

The analysis thread is designed to process approximately 10 frames per second, using a 100 ms interval enforced in `MainActivity.kt`. This reduces unnecessary processing while maintaining sufficient temporal resolution for detecting prolonged eye closures.

No server-side inference or cloud-based scoring is described in the supplied project material.

2 System Architecture

DriveSafe follows a sequential processing pipeline. Each analysed frame passes through image conversion, face landmark extraction, feature calculation, temporal analysis and scoring.

2.1 Camera Input

CameraX provides two use cases: preview and image analysis. The analysis pipeline receives frames in `YUV\_420\_888` format.

The keep-only-latest backpressure strategy prevents old frames from building up when processing is slower than the camera.

An additional time-based gate in `MainActivity.kt` limits analysis to approximately 10 frames per second.

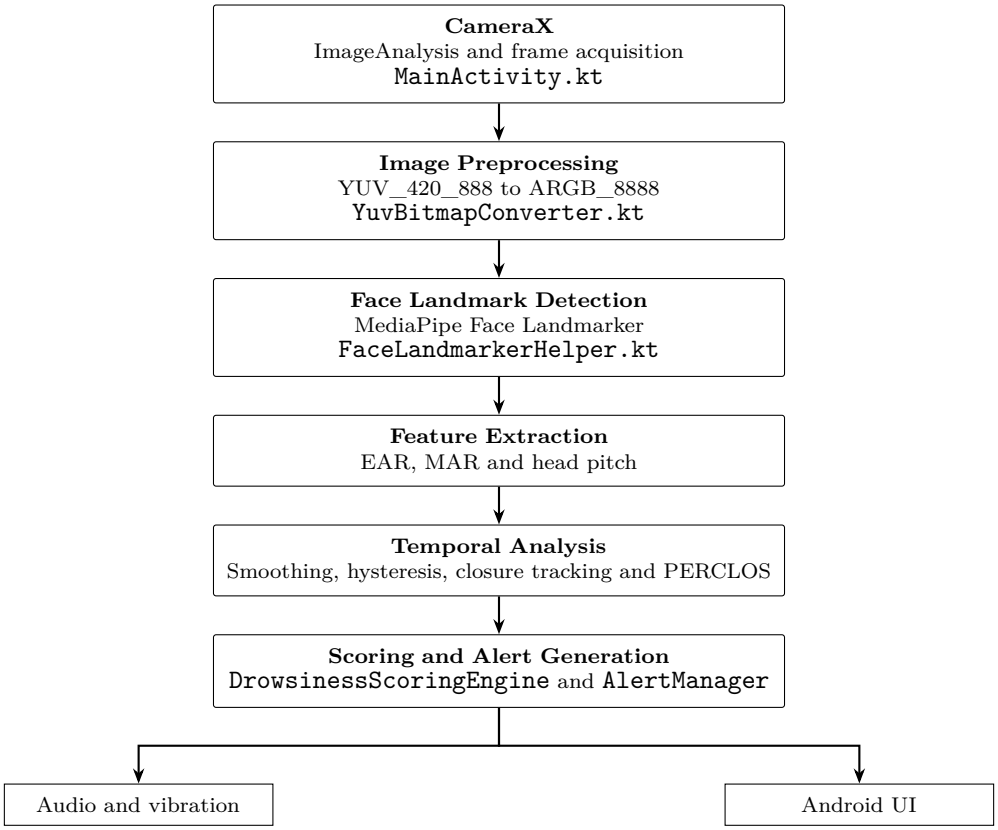


Figure 1: Overall architecture of the DriveSafe detection system.

2.2 Image Preprocessing

Camera frames contain separate luminance and chrominance planes. The custom `YuvBitmapConverter` reads the Y, U and V buffers, accounts for their row and pixel strides, and converts the image into an `ARGB\_8888` bitmap.

The converter also applies the required sensor rotation and reuses a pixel buffer to reduce repeated memory allocation.

2.3 Face Landmark Detection

The converted bitmap is passed to MediaPipe through `landmarker.detectForVideo(mpImage, timestampMs)`.

The task operates in `RunningMode.VIDEO`. During continuous tracking, information from previous frames helps locate the face in subsequent frames. The detector is run again when tracking is lost.

The output consists of normalized facial landmark coordinates.

2.4 Feature Extraction

Selected landmarks are converted into pixel coordinates. The system then calculates:

- Left and right Eye Aspect Ratios.
- Mouth Aspect Ratio.
- Approximate head pitch.

These values are passed to the detection components for temporal analysis.

2.5 Temporal Analysis

The raw measurements are processed using an exponential moving average and dual-threshold hysteresis.

The closure tracker measures how long both eyes remain closed. The PERCLOS calculator maintains a rolling history of valid observation intervals.

Yawn and nod trackers independently monitor sustained changes in mouth opening and head position.

2.6 Scoring and Alerts

The scoring engine combines the extracted indicators into a score between 0 and 100.

The resulting detection state is included in a `DetectionSnapshot`. When an alert is required, `AlertManager` handles audio and vibration separately from the UI update.

3 Computer Vision Foundations

3.1 Face Detection Approaches

Face detection methods have evolved from simple rectangular detection to dense facial landmark estimation.

Table 3: Comparison of face-analysis approaches.

Approach	Method	Characteristics
Haar Cascades	Integral images and AdaBoost	Fast face detection, but sensitive to lighting and head rotation.
Dlib	HOG, SVM and regression trees	Provides facial landmarks but can be computationally demanding on mobile devices.
MediaPipe	BlazeFace and face landmark regression	Provides dense facial landmarks and supports video tracking.

DriveSafe uses MediaPipe Face Landmarker because facial landmarks are required to measure small movements around the eyes and mouth.

3.2 BlazeFace Detector

BlazeFace is a lightweight face detector designed for mobile hardware [5].

It identifies face candidates and facial keypoints, which provide the region of interest for the landmark model. Its main role is face localisation at startup and when tracking is lost.

3.3 Facial Landmark Regression

The landmark model estimates three-dimensional coordinates for points across the face [6].

The coordinates are normalised relative to the image:

$$x, y \in [0, 1]$$

DriveSafe uses the image-plane coordinates for its geometric calculations. The depth coordinate is not used in the supplied detection formulas.

3.4 Video Tracking

MediaPipe uses previous frame information to assist with subsequent face tracking.

The configured minimum tracking confidence in the research is `0.5f`. Tracking reduces the need to repeatedly run the full face detector.

Monotonic timestamps are supplied by the analysis thread for video-mode processing.

4 Mathematical Foundations

This section describes the measurements used by DriveSafe and the equations behind the detection logic.

4.1 Pixel Coordinates and Distance

MediaPipe returns normalised landmark coordinates. For an image with width W and height H, a landmark is converted into pixel coordinates using:

$$p = (xW,\ yH) \tag{1}$$

where x and y are the normalised coordinates.

The Euclidean distance between two landmarks $p_a = (x_a, y_a)$ and $p_b = (x_b, y_b)$ is:

$$d(p_a, p_b) = \sqrt{(x_a - x_b)^2 + (y_a - y_b)^2} \tag{2}$$

Only the two image-plane coordinates are used.

4.2 Eye Aspect Ratio (EAR)

EAR measures the vertical opening of the eye relative to its horizontal width.

Six landmarks are used for each eye: two eye corners, two upper eyelid points and two lower eyelid points.

The formula is:

$$\mathrm{EAR} = \frac{\|p_2 - p_6\| + \|p_3 - p_5\|}{2\|p_1 - p_4\|} \tag{3}$$

A higher EAR generally indicates an open eye. As the eyelids close, the vertical distances decrease and EAR approaches zero.

Typical open-eye values reported in the research are approximately 0.25–0.38.

The default classification thresholds are:

$$\tau_{\text{closed}} = 0.18, \qquad \tau_{\text{open}} = 0.22$$



Open eye
EAR ≈ 0.28

Closed eye
EAR ≈ 0.04

Figure 2: Illustration of the landmark geometry used for EAR.

EAR is a ratio of distances, so uniform changes in image scale cancel out. This makes the measurement less sensitive to how far the driver sits from the camera.

4.3 Left and Right EAR

EAR is calculated independently for each eye, producing EAR_L and EAR_R.

The implementation uses the following calculation:

EyeAspectRatioCalculator.kt

```
val horizontal = distance(points[0], points[3])

val ear = (distance(points[1], points[5]) +
           distance(points[2], points[4])) /
          (2.0 * horizontal)
```

The six points correspond to p_1 through p_6 in Equation 3.

4.4 Mouth Aspect Ratio (MAR)

MAR measures mouth opening using two mouth-corner landmarks and three vertical distances between the upper and lower lips.

$$\text{MAR} = \frac{\|u_C - l_C\| + \|u_L - l_L\| + \|u_R - l_R\|}{2\|c_L - c_R\|} \tag{4}$$

A larger MAR indicates a wider mouth opening.

DriveSafe uses the following conditions for yawn detection:

- MAR greater than 0.65.
- Opening sustained for at least 1500 ms.
- Vertical-to-horizontal ratio of at least 0.15.
- MAR falls below 0.50 before another yawn is counted.
- A cooldown of 6000 ms is applied.

Brief speech-related mouth movements are filtered using the duration requirement.

The formula and thresholds are retained as described in the project research. Since three vertical distances are divided by twice the mouth width, the normalisation should remain unchanged when applying these thresholds.

4.5 Head Pitch Estimation

The current implementation estimates head pitch using three facial landmarks:

- Forehead: index 10.
- Nose tip: index 1.
- Chin: index 152.

The upper and lower distances are:

$$d_{\text{upper}} = \|p_{10} - p_1\| \tag{5}$$

$$d_{\text{lower}} = \|p_1 - p_{152}\| \tag{6}$$

Their ratio is:

$$\rho = \frac{d_{\text{upper}}}{d_{\text{lower}}} \tag{7}$$

The approximate pitch is calculated as:

$$\theta_{\text{pitch}} \approx (1 - \rho) \times 45^\circ \tag{8}$$

A nod is confirmed when the estimated pitch remains below -18° for at least 1200 ms.

This is a two-dimensional approximation and cannot independently distinguish pitch, yaw and roll.

4.6 Exponential Moving Average

Raw EAR measurements can fluctuate because of landmark instability. DriveSafe applies an exponential moving average before classifying eye state.

$$\overline{\text{EAR}}_t = \alpha \text{EAR}_t + (1 - \alpha)\overline{\text{EAR}}_{t-1} \tag{9}$$

The smoothing factor used in the research is:

$$\alpha = 0.50$$

This reduces the effect of individual noisy measurements while allowing genuine changes to be detected.

4.7 Dual-Threshold Hysteresis

Using a single threshold can cause rapid switching when EAR fluctuates around that value.

DriveSafe instead uses two thresholds:

$$s_t = \begin{cases} \text{closed}, & \overline{\text{EAR}}_t \leq \tau_{\text{closed}} \\ \text{open}, & \overline{\text{EAR}}_t \geq \tau_{\text{open}} \\ s_{t-1}, & \text{otherwise} \end{cases} \tag{10}$$

The default values are 0.18 and 0.22 respectively. Between these values, the previous classification is retained.

4.8 PERCLOS

PERCLOS represents the proportion of valid observation time during which the eyes are closed.

DriveSafe calculates it using elapsed time intervals rather than frame counts:

$$\text{PERCLOS} = \frac{\sum_{i \in C} \Delta t_i}{\sum_{j \in V} \Delta t_j} \tag{11}$$

where:

- C represents valid closed-eye intervals.
- V represents all valid observation intervals.
- Δt represents the duration of each interval.

The implementation maintains a rolling 30-second window.

A minimum of 5 seconds of valid observation time is required before a value is returned. Intervals longer than 500 ms are excluded to prevent large gaps from distorting the result.

Since DriveSafe uses EAR-based bilateral closure rather than the original eyelid-coverage definition, its measurement is more accurately described as PERCLOS-style.

4.9 Drowsiness Sub-Scores

The composite score uses normalised sub-scores capped at 100.

$$S_{\text{sustained}} = \min\left(100, \frac{\Delta t_{\text{closure}}}{2000} \times 100\right) \tag{12}$$

$$S_{\text{perclos}} = \min(100, \text{PERCLOS} \times 250) \tag{13}$$

$$S_{\text{freq}} = \min(100, N_{\text{prolonged},45s} \times 25) \tag{14}$$

Here, $\Delta t_{\text{closure}}$ is the current continuous bilateral closure duration, and $N_{\text{prolonged},45s}$ is the number of prolonged closure episodes in the last 45 seconds.

5 Drowsiness Detection Logic

5.1 Eye Classification

`EyeClassifier` processes each eye independently.

For every frame, it:

1. Applies EMA smoothing to the raw EAR.
2. Compares the result with the two thresholds.
3. Retains the previous state inside the hysteresis band.

5.2 Bilateral Eye Fusion

A one-sided eye closure can occur during winking, mirror checking or landmark detection errors.

To reduce false detections, DriveSafe combines both eye states. A closure is considered bilateral only when both eyes are classified as closed.

Table 4: Bilateral eye fusion rules.

Left eye	Right eye	Fused state	Interpretation
Closed	Closed	Closed	Both eyes agree.
Open	Open	Open	Normal eye state.
Closed	Open	Open	Asymmetric closure.
Open	Closed	Open	Asymmetric closure.

The fused state is passed to the closure tracker and PERCLOS calculator.

5.3 Blink and Closure Classification

`ClosureEpisodeTracker` measures the duration for which both eyes remain closed.

Table 5: Closure duration categories.

Category	Duration	System action
Normal blink	Below 200 ms	Ignored.
Monitor	200–500 ms	Slow blink or eye flutter.
Potential drowsiness	500–1000 ms	Prolonged episode counted.
Serious drowsiness	1000–2000 ms	At least drowsy state.
Critical closure	2000 ms or more	Critical state override.

The tracker also includes anti-flicker bridging. If the eyes reopen for less than 250 ms, the closure episode is not immediately ended.

This prevents brief eyelid movements from repeatedly resetting the closure timer.

5.4 Microsleep Detection

The project does not describe a separate microsleep flag. Instead, prolonged closures are handled through the duration categories.

A bilateral closure of 1000 ms or more forces at least the drowsy state. A closure of 2000 ms or more immediately forces the critical state, regardless of the composite score.

5.5 Yawn Detection

The yawn tracker uses MAR to identify sustained mouth opening.

A yawn is confirmed when:

- MAR exceeds 0.65.
- The condition lasts for at least 1500 ms.
- The vertical-to-horizontal ratio is at least 0.15.

After detection, MAR must fall below 0.50. A 6000 ms cooldown prevents repeated detection of the same yawn.

A confirmed yawn contributes an additional score bonus.

5.6 Head Nod Detection

The head-pose estimator monitors the facial midline ratio and calculates approximate pitch using Equation 8.

A nod is confirmed when pitch remains below -18° for at least 1200 ms.

This duration requirement helps distinguish sustained head drops from ordinary glances at the instrument panel.

5.7 Rolling PERCLOS

The `PerclosCalculator` maintains a rolling collection of observation samples in an `ArrayDeque`.

Samples older than 30 seconds are removed. Observation intervals exceeding 500 ms are discarded, and a value is returned only after at least 5 seconds of valid data has accumulated.

5.8 Face Visibility Handling

Missing facial landmarks must not be interpreted as closed eyes.

The visibility tracker requires three consecutive successful detections before the face is considered locked.

After locking, it allows a grace period of 1500 ms for temporary face loss.

Observation gaps exceeding 600 ms trigger `breakContinuity()`, preventing missing time from being counted as closed-eye time.

6 Implementation and Runtime

6.1 MainActivity.kt

`MainActivity` manages camera initialisation, frame acquisition, processing and UI updates.

It uses a single-thread executor for analysis and applies two processing guards:

- A minimum interval of 100 ms between inferences.
- An atomic flag using `processingFrame.compareAndSet(false, true)` to prevent overlapping frame processing.

After analysis, the resulting snapshot is passed to the alert manager and the UI.

6.2 FaceLandmarkerHelper.kt

This class manages the MediaPipe Face Landmarker task in `RunningMode.VIDEO`.

It converts the result into pixel-space coordinates and extracts the required landmark sets for the eyes, mouth and head-pose calculation.

The coordinate conversion is:

```
Point2D(
    p.x().toDouble() * width,
    p.y().toDouble() * height
)
```

6.3 Detection Components

The main detection classes have the following roles.

Table 6: Responsibilities of the detection components.

Component	Responsibility
`EyeAspectRatioCalculator`	Calculates Euclidean distances and EAR.
`MouthAspectRatioCalculator`	Calculates MAR and supplies yawn detection measurements.
`EyeClassifier`	Applies smoothing, hysteresis and bilateral fusion.
`ClosureEpisodeTracker`	Tracks closure duration and episode categories.
`PerclosCalculator`	Maintains rolling valid and closed-eye intervals.
`HeadPoseEstimator`	Calculates approximate head pitch and nod events.
`DrowsinessDetector`	Coordinates the detection components and creates snapshots.
`DrowsinessScoringEngine`	Calculates the composite score and detection state.

6.4 YUV-to-RGB Conversion

`YuvBitmapConverter` reads the Y, U and V planes using their respective row and pixel strides.

The conversion uses fixed-point integer arithmetic:

```
val y = (yBuffer.get(yIndex).toInt() and 0xff) - 16
val u = (uBuffer.get(uIndex).toInt() and 0xff) - 128
val v = (vBuffer.get(vIndex).toInt() and 0xff) - 128

val c = y.coerceAtLeast(0)

val red =
    ((298 * c + 409 * v + 128) shr 8)
        .coerceIn(0, 255)

val green =
    ((298 * c - 100 * u - 208 * v + 128) shr 8)
        .coerceIn(0, 255)

val blue =
    ((298 * c + 516 * u + 128) shr 8)
        .coerceIn(0, 255)

scratchPixels[row * width + col] =
    (0xff shl 24) or
    (red shl 16) or
    (green shl 8) or blue
```

The coefficients are scaled by 256. The right shift divides the result by 256, while the added value 128 provides rounding.

The resulting integer represents an `ARGB\_8888` pixel with full alpha. The converted bitmap is rotated to match the camera orientation.

6.5 Frame Execution Sequence

Each analysed frame follows this sequence:

1. CameraX supplies an `ImageProxy`.
2. The frame is converted into an ARGB bitmap.
3. MediaPipe extracts facial landmarks.
4. EAR, MAR and head pitch are calculated.
5. The detector performs temporal tracking.
6. The scoring engine determines the state.
7. An alert is dispatched when required.
8. The UI is updated using the detection snapshot.

6.6 Thread Handling

Frame analysis runs on `Executors.newSingleThreadExecutor()`.

Using a single analysis thread preserves frame order, which is important for EMA smoothing, hysteresis, closure tracking and PERCLOS.

UI updates are posted to the main thread using `runOnUiThread`. Audio and vibration are handled through the alert manager and Android system services.

6.7 Frame Throttling

Two mechanisms limit processing load:

- CameraX uses `STRATEGY\_KEEP\_ONLY\_LATEST`.
- The application enforces a minimum 100 ms interval between inference calls.

This gives a target analysis rate of approximately 10 frames per second.

6.8 Alert Dispatch

When `alarmRequested` is set in the `DetectionSnapshot`, `MainActivity` calls `AlertManager.playAlert(level)`. The manager selects the appropriate alert level, plays audio using the alarm audio stream and requests vibration.

6.9 UI Updates

The detection snapshot contains the state, score, EAR, MAR, pitch and relevant flags.

These values are displayed through `MainActivity.render(snapshot)` on the UI thread.

7 Drowsiness Scoring and Alert Escalation

7.1 Composite Score

DriveSafe combines multiple indicators into a score between 0 and 100.

$$\begin{aligned} S = {} & 0.40 S_{\text{sustained}} + 0.25 S_{\text{perclos}} \\ & + 0.20 S_{\text{freq}} + 0.15 S_{\text{avg}} \\ & + B_{\text{yawn}} + B_{\text{nod}} \end{aligned} \tag{15}$$

The weights represent the relative contributions of the indicators.

Table 7: Scoring contributions.

Term	Weight	Input	Normalisation
$S_{\text{sustained}}$	0.40	Current bilateral closure	$\Delta t/2000 \times 100$
S_{perclos}	0.25	30-second PERCLOS	PERCLOS $\times$ 250
S_{freq}	0.20	Prolonged closures in 45 seconds	Count $\times 25$
S_{avg}	0.15	Average eye openness	Definition not supplied
B_{yawn}	Additive	Confirmed yawn	Magnitude not supplied
B_{nod}	Additive	Confirmed nod	Magnitude not supplied

The research does not specify the calculation for S_{avg}, the magnitudes of the two bonuses or the numerical score boundaries for the detection states.

7.2 Warning and Critical Conditions

The final detection state depends on the composite score and the duration of the current eye closure.

The following duration overrides are used:

- Closure of at least 1000 ms: the state is forced to at least drowsy.
- Closure of at least 2000 ms: the state is forced to critical.

These overrides apply regardless of the score. A face-visibility failure produces a face-loss state instead of a drowsiness classification.

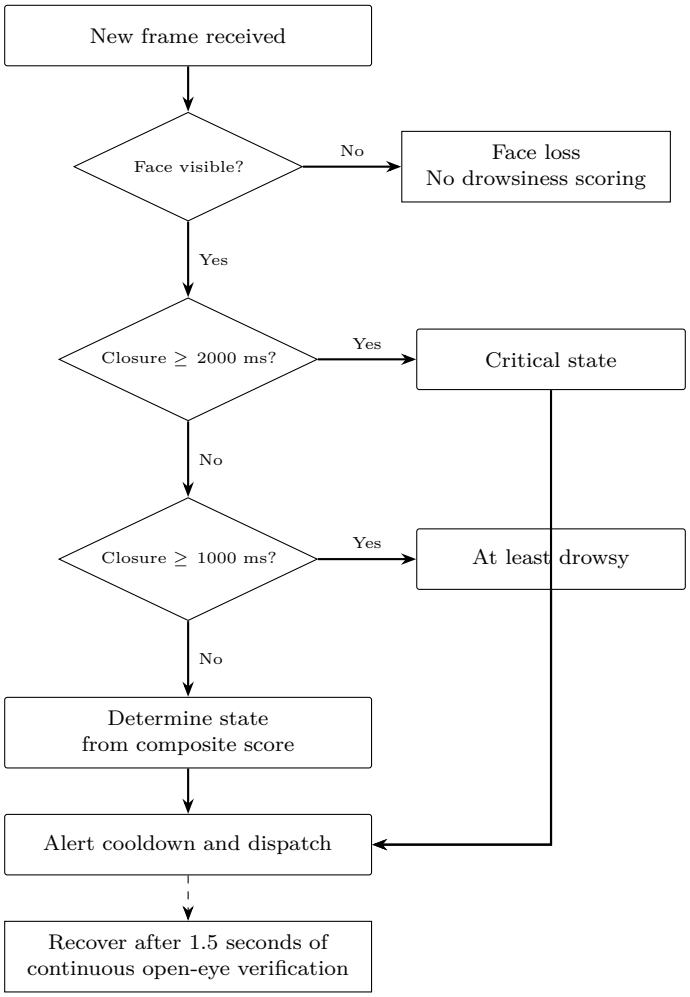


Figure 3: Decision flow from frame observation to alert generation.

7.3 Alert Cooldown

Cooldowns prevent the alarm from restarting on every qualifying frame.

The configured intervals are:

- Warning alert: 4 seconds.
- Critical alert: 5 seconds.

7.4 Recovery Conditions

An active alarm is cleared after 1.5 seconds of continuous open-eye verification.

This is separate from the 250 ms recovery confirmation used by the closure tracker.

The shorter interval ends a closure episode, whereas the longer interval determines when an active alarm can be cleared.

8 Real-World Challenges and Limitations

Camera-based driver monitoring is affected by lighting, occlusion, head movement and individual facial differences.

Table 8: Challenges and current handling.

Condition	Effect	Current handling
Low light	Noise and motion blur reduce landmark reliability.	EMA and hysteresis reduce fluctuations, but the system still depends on image quality.
Glasses and reflections	Eye landmarks may be partially hidden or displaced.	Bilateral fusion reduces some one-sided errors. Two-sided reflections remain a limitation.
Head rotation	Changes the projected eye geometry and may affect EAR.	Duration checks and bilateral fusion help reduce false alarms. Full 3D compensation is proposed.
Facial variation	Different eye shapes produce different baseline EAR values.	Per-driver calibration is described in the research.
Dropped frames	Large observation gaps affect temporal measurements.	PERCLOS uses elapsed time and excludes invalid intervals.
Landmark instability	Small changes in landmark positions affect measurements.	EMA smoothing, hysteresis and flicker bridging.

Two limitations are particularly important.

First, bilateral fusion ignores genuine one-sided closures. This reduces false alarms but means the system does not detect unilateral eye problems.

Second, the current head-pose measurement is two-dimensional. Large head rotations can change the measured ratios even when eye openness remains unchanged.

The supplied research does not provide field-validation results, clinical evaluation or measured detection accuracy. Therefore, no accuracy or performance figures are claimed.

9 Future Improvements

The following improvements are described in the project research. Implemented features and proposed changes are listed separately.

Table 9: Implementation status of improvements.

Improvement	Status	Details
Per-driver calibration	Described as implemented	`CalibrationManager.kt`
Adaptive baseline tracking	Described as implemented	`AdaptiveBaselineTracker.kt`
3D head-pose estimation	Proposed	Perspective-n-Point approach
EAR compensation	Proposed	Compensation for head rotation
Near-infrared illumination	Proposed	Requires additional hardware

9.1 Adaptive Calibration

A fixed EAR threshold may not work equally well for all drivers because natural eye shape varies.

The research describes a calibration phase lasting approximately 3–5 seconds. During this period, the system records the 90th-percentile open-eye EAR as the baseline.

The personalised thresholds are:

$$\tau_{\text{closed}} = 0.60 \cdot \text{EAR}_{\text{baseline}} \tag{16}$$

$$\tau_{\text{open}} = 0.75 \cdot \text{EAR}_{\text{baseline}} \tag{17}$$

The adaptive baseline is also described as slowly changing with a factor of $\beta = 0.001$, provided the driver is verified to be awake, has a level head and has open eyes.

The exact baseline update expression is not included in the supplied research.

9.2 3D Head Pose Estimation

The current pitch estimate uses a two-dimensional landmark ratio. It cannot distinguish pitch from simultaneous yaw and roll.

A proposed improvement is Perspective-n-Point (PnP) estimation using known three-dimensional facial points, their image positions and camera intrinsics.

The projection relationship is:

$$s \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = K \begin{bmatrix} R & t \end{bmatrix} \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix} \tag{18}$$

where K is the camera intrinsic matrix, R is rotation, t is translation and s is the projective scale factor.

The solved rotation can provide yaw, pitch and roll.

9.3 EAR Compensation

The research also proposes compensating EAR for head pitch:

$$\text{EAR}_{\text{compensated}} = \frac{\text{EAR}_{\text{measured}}}{\cos \theta_{\text{pitch}}} \tag{19}$$

This is a heuristic proposal and has not been validated. It assumes pitch affects the vertical eye distance in a predictable way, which may not hold under combined head rotation.

It should be tested using recorded data before implementation.

9.4 Near-Infrared Illumination

Standard RGB cameras may struggle in poorly illuminated vehicle cabins.

A possible hardware improvement is an NIR-sensitive camera with 940 nm illumination. This can improve facial visibility in low-light environments.

This requires additional hardware and cannot be implemented through software alone on a standard phone front camera.

10 Technical Reference

10.1 Numerical Thresholds

Table 10: Main detection thresholds.

Parameter	Value	Purpose
Closed-eye EAR	0.18	Eye closure classification.
Open-eye EAR	0.22	Eye reopening classification.
EMA factor α	0.50	EAR smoothing.
Calibrated closed threshold	$0.60 \times \text{EAR}_{baseline}$	Personalised closure threshold.
Calibrated open threshold	$0.75 \times \text{EAR}_{baseline}$	Personalised reopening threshold.
Adaptive factor β	0.001	Baseline update.
Yawn threshold	MAR > 0.65	Mouth opening.
Yawn reopening	MAR < 0.50	Re-arm after yawn.
Vertical-to-horizontal ratio	0.15	Yawn shape condition.
Nod pitch	Below $-18°$	Head nod.
Tracking confidence	0.5f	Face tracking threshold.

10.2 Landmark Indices

The following MediaPipe landmark indices are used for feature extraction.

Table 11: Facial landmark indices used by DriveSafe.

Group	Role	Indices
Left eye	$p_1, p_2, p_3, p_4, p_5, p_6$	33, 160, 158, 133, 153, 144
Right eye	$p_1, p_2, p_3, p_4, p_5, p_6$	362, 385, 387, 263, 373, 380
Mouth corners	c_L, c_R	61, 291
Mouth centre	u_C, l_C	13, 14
Mouth left	u_L, l_L	82, 87
Mouth right	u_R, l_R	312, 317
Head midline	Forehead, nose, chin	10, 1, 152

10.3 Timing Parameters

Table 12: Temporal parameters used by DriveSafe.

Parameter	Value	Purpose
Frame analysis interval	100 ms	Minimum interval between inference calls.
Normal blink	Below 200 ms	Normal eye closure.
Monitor closure	200–500 ms	Identifies prolonged blinking.
Prolonged closure	500–1000 ms	Indicates possible drowsiness.
Drowsiness override	1000 ms	Forces at least the drowsy state.
Critical closure	2000 ms	Forces the critical state.
Closure recovery	250 ms	Confirms the end of a closure episode.
Yawn duration	1500 ms	Minimum sustained mouth opening.
Nod duration	1200 ms	Minimum duration for nod detection.
Yawn cooldown	6000 ms	Prevents repeated yawn detection.
PERCLOS window	30 s	Rolling observation period.
Closure frequency window	45 s	Counts prolonged closure episodes.
Maximum valid interval	500 ms	Excludes large PERCLOS intervals.
Continuity break interval	600 ms	Resets observation continuity.
Face-loss grace period	1500 ms	Allows temporary face disappearance.
Face lock requirement	3 frames	Consecutive detections required for face lock.
Warning cooldown	4 s	Minimum interval between warning alerts.
Critical cooldown	5 s	Minimum interval between critical alerts.
Alarm recovery	1.5 s	Continuous open-eye time required to clear an alarm.
Calibration period	3–5 s	Initial driver calibration.

10.4 Composite Scoring Weights

The overall drowsiness score combines eye-closure duration, PERCLOS, closure frequency and additional behavioural indicators.

Table 13: Contributions to the composite drowsiness score.

Component	Weight	Description
$S_{\text{sustained}}$	0.40	Current bilateral eye-closure duration.
S_{perclos}	0.25	Proportion of valid observation time with closed eyes.
S_{freq}	0.20	Number of prolonged closures in the last 45 seconds.
S_{avg}	0.15	Average eye-openness contribution.
B_{yawn}	Additive	Bonus for a confirmed yawn.
B_{nod}	Additive	Bonus for a confirmed head nod.

10.5 Detection States

Eye classification and overall drowsiness classification are separate parts of the detection pipeline.

10.5.1 Eye Classification

Table 14: Eye classification based on smoothed EAR.

State	EAR Condition	Description
Open	EAR ≥ 0.22	Eye is classified as open.
Closed	EAR ≤ 0.18	Eye is classified as closed.
Previous state	$0.18 <$ EAR < 0.22	Previous classification is retained.

10.5.2 Drowsiness Classification

Table 15: Overall detection states.

State	Condition
Normal	Lowest drowsiness score range.
Monitor	Intermediate drowsiness score range.
Warning / Drowsy	Warning score condition or eye closure of at least 1000 ms.
Critical	Critical score condition or eye closure of at least 2000 ms.
Face Loss	Face tracking unavailable beyond the allowed grace period.

10.6 Implementation Notes

The following details should be considered when interpreting the implementation:

- The definition of S_{avg}, the magnitudes of the yawn and nod bonuses, and the numerical score boundaries are not specified in the available research.
- Calibration and adaptive-baseline functionality are described in the research, but their source files are absent from the reviewed file list.
- The PERCLOS implementation uses separate 500 ms and 600 ms checks for interval validity and continuity.
- The research refers to both 468 and 478 facial landmarks. All indices used by DriveSafe are below 468.
- No measured accuracy, performance benchmarks or real-world validation results are available.

11 Conclusion

DriveSafe converts facial geometry into a time-aware driver drowsiness detection system.

Its main design choices address the limitations of single-frame detection. EAR is smoothed and classified using hysteresis, both eyes must agree before a closure is counted, and closure duration is tracked to distinguish normal blinks from prolonged eye closure.

The system also monitors yawning, head movement and rolling PERCLOS. These indicators are combined into a drowsiness score, with duration-based overrides and audio and vibration alerts.

The main areas requiring further work are the incomplete scoring parameters, verification of the calibration components, improved head-pose estimation and real-world testing.

Future development should include unit testing of the detection thresholds, collection of labelled driving data and evaluation under different lighting and head-pose conditions.

12 References and Further Reading

References

[1] P. Viola and M. Jones, "Rapid Object Detection using a Boosted Cascade of Simple Features," *Proceedings of IEEE CVPR*, 2001.

[2] N. Dalal and B. Triggs, "Histograms of Oriented Gradients for Human Detection," *Proceedings of IEEE CVPR*, 2005.

[3] V. Kazemi and J. Sullivan, "One Millisecond Face Alignment with an Ensemble of Regression Trees," *Proceedings of IEEE CVPR*, 2014.

[4] D. E. King, "Dlib-ml: A Machine Learning Toolkit," *Journal of Machine Learning Research*, vol. 10, pp. 1755–1758, 2009.

[5] V. Bazarevsky *et al.*, "BlazeFace: Sub-millisecond Neural Face Detection on Mobile GPUs," arXiv:1907.05047, 2019.

[6] Y. Kartynnik *et al.*, "Real-time Facial Surface Geometry from Monocular Video on Mobile GPUs," arXiv:1907.06724, 2019.

[7] T. Soukupová and J. Čech, "Real-Time Eye Blink Detection using Facial Landmarks," *21st Computer Vision Winter Workshop*, 2016.

[8] D. F. Dinges and R. Grace, "PERCLOS: A Valid Psychophysiological Measure of Alertness as Assessed by Psychomotor Vigilance," FHWA Tech Brief FHWA-MCRT-98-006, 1998.

[9] Google, "MediaPipe Face Landmarker Guide." `https://ai.google.dev/edge/mediapipe/solutions/vision/face_landmarker`

[10] Android Developers, "CameraX Overview." `https://developer.android.com/media/camera/camerax`